APMA HW5

1. For d-dim 1, simple random walk,

$P(\xi_i = 1) = \frac{1}{2}$, $P(\xi_i = -1) = \frac{1}{2}$.

$X_n = i$

$P(X_{n+1} = i+1 \mid X_n = i) = P(\xi_{n+1} = 1) = \frac{1}{2}$

$P(X_{n+1} = i-1 \mid X_n = i) = P(\xi_{n+1} = -1) = \frac{1}{2}$

From the definition of 1-dim simple random walk,

$$X_n(w) = X_0 + \sum_{i=1}^{n} \xi_i(w).$$

ξ_{n+1} can be either +1 or -1, and it is independent of previous states given ~~it is iid.~~ ξ_i is iid. Again, say $X_n = i$, ~~[redacted]~~ whether the probability of next state is $i+1$ or $i-1$ depends solely on the current state but not any preceding sequences. Therefore, the 1-dim simple random walk is a markov chain:

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \cdots X_0 = i_0) = P(X_{n+1} = j \mid X_n = i)$$

A MC is homogeneous if it's transitional probabilities is stable and doesn't change over time. We know

$$P(\xi_i = +1) = P(\xi_i = -1) = \frac{1}{2}.$$

At any time, the transition probabilities doesn't depend on time n, therefore the 1-dim simple random walk is indeed an homogenous markov chain.

The transition matrix is the following:

$$P = \begin{pmatrix} \ddots & \vdots & \vdots & \vdots & \vdots & \\ \cdots & 0 & \frac{1}{2} & 0 & 0 & 0 & \cdots \\ \cdots & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & \cdots \\ \cdots & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & \cdots \\ \cdots & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & \cdots \\ \cdots & 0 & 0 & 0 & \frac{1}{2} & 0 & \cdots \end{pmatrix}$$

2. Theorem 1.1 states that the state y is recurrent if and only if $\sum_{n=1}^{\infty} P(X_n = y \mid X_0 = y) = \infty$.

We have to show the above with $y = 0$.

For 1-dim simple random walk, the probability returns to origin after $2n$ steps are $P(X_{2n} = 0 \mid X_0 = 0) = \binom{2n}{n}\left(\frac{1}{2}\right)^{2n}$

Also, the walker needs to go n steps left n steps right exactly.

$P(X_{2n+1} = 0 \mid X_0 = 0) = 0$ (The walker can't get back to the origin with odd number steps)

The sum need to calculate:

$$\sum_{n=1}^{\infty} P(X_{2n} = 0 \mid X_0 = 0) = \sum_{n=1}^{\infty} \binom{2n}{n}\left(\frac{1}{2}\right)^{2n}$$

Stirling Formula:

$$\lim_{n\to\infty} \frac{n!}{n^n \cdot e^{-n}\sqrt{2\pi n}} = 1 \Rightarrow n! \sim \left(\frac{n}{e}\right)^n \cdot \sqrt{2\pi n} \quad (1)$$

$\binom{2n}{n} = \frac{(2n)!}{n!\,n!}$ (2), plug (1) to (2), we get $\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}}$,

put it back to the sum:

$$\sum_{n=1}^{\infty} \frac{4^n}{\sqrt{\pi n}}\left(\frac{1}{2}\right)^{2n} = \sum_{n=1}^{\infty} \frac{1}{\sqrt{\pi n}} > \sum_{n=1}^{\infty} \frac{1}{n}$$

↑ Divergence

Therefore, the sum we need is also divergent.

So we prove the point $0 \in \mathbb{Z}$ is recurrent for 1-dim SRW.

$$\pi = P^T \pi$$

Acolumn Vector

$S \times S$ dim Matrix

$S \times 1$ dim Vector

$$\begin{pmatrix} \pi(x_1) \\ \pi(x_2) \\ \vdots \\ \pi(x_S) \end{pmatrix} = \begin{pmatrix} P(x_1, x_1) & P(x_2, x_1) & \cdots & P(x_S, x_1) \\ P(x_1, x_2) & & & \\ \vdots & & & \\ P(x_1, x_S) & & & P(x_S, x_S) \end{pmatrix} \begin{pmatrix} \pi(x_1) \\ \pi(x_2) \\ \vdots \\ \pi(x_S) \end{pmatrix}$$

$$= \sum_{i=1}^{S} P(x_i, x_j)\pi(x_i) = \pi(x_j), \quad \forall j \in \{1, \cdots S\}$$

y, x sustitute x_i, x_j $\Rightarrow$ $\sum_{\text{yes}} P(y, x)\pi(y) = \pi(x), \quad \forall x \in X$